

Meaningful Vector Problems for Chemistry

Dot product, cross product, normalization, projections, angles, torque, and molecular geometry

Student handout

Purpose. These problems use vectors in chemically meaningful ways: forces on atoms, displacements during bond distortion, bond vectors, molecular-plane normals, dipole addition, torque, Lorentz force on ions, reaction-coordinate directions, and stereochemical volume. The point is not to multiply random chemical formulas. The point is to use vector operations where direction has physical or molecular meaning.

Unit conversions used in this handout.

$$1 \text{ pN} = 10^{-12} \text{ N}, \quad 1 \text{ \AA} = 10^{-10} \text{ m}, \quad 1 \text{ pN \AA} = 10^{-22} \text{ J}$$

$$N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1}$$

Vector quick reference.

Operation	Formula	Chemical use
Magnitude	$ \mathbf{v} = \sqrt{v_x^2 + v_y^2 + v_z^2}$	Bond length, displacement length, force strength
Unit vector	$\hat{\mathbf{v}} = \mathbf{v}/ \mathbf{v} $	Direction of a bond, force, normal, or reaction coordinate
Dot product	$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z$	Work, projection, angle between bonds
Angle	$\cos \theta = (\mathbf{a} \cdot \mathbf{b})/(\mathbf{a} \mathbf{b})$	Bond angles, alignment of force and motion
Projection	$a_{\parallel} = \mathbf{a} \cdot \hat{\mathbf{b}}$	Force along a bond or reaction coordinate
Cross product	$\mathbf{a} \times \mathbf{b}$	Torque, plane normal, Lorentz force
Scalar triple product	$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$	Signed volume; stereochemical handedness test

Problem 1. Bond vector, bond length, and normalization

In a small-molecule structure, the carbon atom is at

$$\mathbf{r}_C = (0.00, 0.00, 0.00)\text{\AA}$$

and the chlorine atom is at

$$\mathbf{r}_{Cl} = (1.10, 1.10, 0.20)\text{\AA}.$$

- a) Find the bond vector from C to Cl.
- b) Find the C–Cl bond length.
- c) Normalize the bond vector to obtain the unit vector pointing from C to Cl.

Problem 2. Dot product as molecular work

In a molecular simulation of water, an external pulling force is applied to one H atom to slightly stretch an O–H bond. At one instant, the force on the H atom is

$$\mathbf{F} = (120\mathbf{i} - 40\mathbf{j} + 0\mathbf{k}) \text{ pN}.$$

During a short time interval, the H atom undergoes the displacement

$$\Delta\mathbf{r} = (0.020\mathbf{i} - 0.010\mathbf{j} + 0\mathbf{k}) \text{ \AA}.$$

- Calculate the work done in pN Å.
- Convert the result to J per molecule.
- Convert the result to kJ/mol.
- Interpret the sign of the work.

Problem 3. Projection of a force along a bond

Use the C–Cl unit vector from Problem 1:

$$\hat{\mathbf{r}}_{C \rightarrow Cl} = (0.701, 0.701, 0.128).$$

A force on the chlorine atom is

$$\mathbf{F} = (90\mathbf{i} + 30\mathbf{j} - 20\mathbf{k}) \text{ pN}.$$

- a) Find the scalar component of the force along the C–Cl bond.
- b) Find the vector component of the force parallel to the C–Cl bond.
- c) Find the magnitude of the force perpendicular to the bond.
- d) Does the force mostly stretch the bond or bend it sideways?

Problem 4. Dot product as a bond angle: the H–O–H angle

A simplified water geometry places O at the origin and the two H atoms at

$$\mathbf{r}_1 = (0.958, 0.000, 0.000)\text{\AA}, \quad \mathbf{r}_2 = (-0.240, 0.927, 0.000)\text{\AA}.$$

- a) Use the dot product to calculate the H–O–H bond angle.
- b) Explain why the dot product is the correct operation here.

Problem 5. Cross product as the normal to a molecular plane

Use the same two O–H bond vectors from Problem 4:

$$\mathbf{r}_1 = (0.958, 0.000, 0.000)\text{\AA}, \quad \mathbf{r}_2 = (-0.240, 0.927, 0.000)\text{\AA}.$$

- a) Calculate $\mathbf{n} = \mathbf{r}_1 \times \mathbf{r}_2$.
- b) Normalize $\mathbf{n}$ to get the unit normal vector to the molecular plane.
- c) What does the direction of this normal vector tell you?

Problem 6. Torque from a force on an atom

A force is applied to an atom in a rotating molecular fragment. Relative to the chosen rotation center, the atom's position vector is

$$\mathbf{r} = (1.20\mathbf{i} - 0.50\mathbf{j} + 0.30\mathbf{k})\text{\AA}.$$

The force on the atom is

$$\mathbf{F} = (0\mathbf{i} + 80\mathbf{j} + 50\mathbf{k})\text{ pN}.$$

- Calculate the torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ in pN $\text{\AA}$.
- Find $|\boldsymbol{\tau}|$.
- Convert the torque magnitude to kJ/mol as an energy scale.
- Explain why torque requires a cross product, not a dot product.

Problem 7. Vector addition of bond dipoles

For a simplified water molecule, use the same two O–H unit bond vectors from Problem 4. Let each O–H bond contribute a vector of magnitude 1.50 D in the direction of the corresponding O–H bond:

$$\hat{\mathbf{r}}_1 = (1.000, 0.000, 0.000), \quad \hat{\mathbf{r}}_2 = (-0.251, 0.968, 0.000).$$

The two bond vectors are therefore

$$\mu_1 = 1.50\hat{\mathbf{r}}_1, \quad \mu_2 = 1.50\hat{\mathbf{r}}_2.$$

- a) Calculate the resultant vector $\mu = \mu_1 + \mu_2$.
- b) Calculate $|\mu|$.
- c) Explain why the two bond contributions do not cancel in water.

Problem 8. Cross product in a mass spectrometer: Lorentz force

A singly charged positive ion moves through a magnetic field. Its velocity is

$$\mathbf{v} = (2.0 \times 10^5 \mathbf{i} + 1.0 \times 10^5 \mathbf{j} + 0 \mathbf{k}) \text{ m/s},$$

and the magnetic field is

$$\mathbf{B} = (0 \mathbf{i} + 0 \mathbf{j} + 0.80 \mathbf{k}) \text{ T}.$$

The ion has charge

$$q = +1.602 \times 10^{-19} \text{ C}.$$

- a) Calculate the magnetic force $\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$.
- b) Find the force magnitude.
- c) State the direction of the deflection in the xy -plane.

Problem 9. Normalized reaction coordinate and force projection

During a small reaction-coordinate step, three internal coordinates change by

$$\Delta \mathbf{q} = (0.050, -0.020, 0.100) \text{Å}.$$

The generalized force in the same coordinate basis is

$$\mathbf{F}_q = (60, -30, 90) \text{ pN}.$$

- a) Normalize $\Delta \mathbf{q}$ to obtain the unit reaction-coordinate direction.
- b) Project $\mathbf{F}_q$ onto this unit direction.
- c) Interpret whether the force is well aligned with the reaction-coordinate step.

Problem 10. Scalar triple product and stereochemical volume

At a tetrahedral carbon, take three bond vectors from carbon to three ligands:

$$\mathbf{a} = (1.20, 0.20, 0.10)\text{\AA},$$

$$\mathbf{b} = (-0.30, 1.10, 0.20)\text{\AA},$$

$$\mathbf{c} = (-0.20, -0.40, 1.00)\text{\AA}.$$

- Compute the scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$.
- Compute the volume of the tetrahedron formed by the carbon atom and the three ligand positions.
- What happens to the sign if two ligand labels are swapped?

Closing principle

A good chemistry vector problem should satisfy two conditions:

1. Each vector must represent a real directed quantity: a force, displacement, bond vector, field, dipole contribution, velocity, or coordinate direction.
2. The vector operation must answer a chemically meaningful question: work, angle, alignment, bending vs stretching, plane orientation, torque, deflection, or handedness.

When those conditions are met, the mathematics is not decorative. It becomes the language of molecular geometry, molecular motion, and molecular energy.